AUTOMATIC IRRIGATION SYSTEM USING IC 555 AND LM741

Course: Analog Electronics

Group No. 59

Group Members :

1. SURAJ SALIAN	2018AAPS0253G
2. AMIT KSHIRSAGAR	2018AAPS0381G
3. AMIT HEGDE	2018AAPS0421G
4. AYUSH VIJ	2018A8PS0371G
5. MITANSH SINGLA	2018A3PS0432G

BITS Pilani
K K Birla Goa Campus

Outline

1. Introduction
2. Design
3. Working of circuit
    - Automatic Water level monitoring system
    - Soil moisture testing unit
    - Relay switch circuit
4. Simulation Results
5. Applications/Advantages
6. Conclusion

Introduction

- The objective of the project is to build a plant irrigation setup using the combination of water level controller and soil moisture tester.

- The unit can regulate the water level in the supply tanks and maintain the required moisture content of the soil. Thus, forming a single unit automatic irrigation system.

- The Automatic Plant Irrigation setup consists of two main blocks: -
  1. Automatic Water Level Monitoring system
  2. Soil Moisture Tester unit

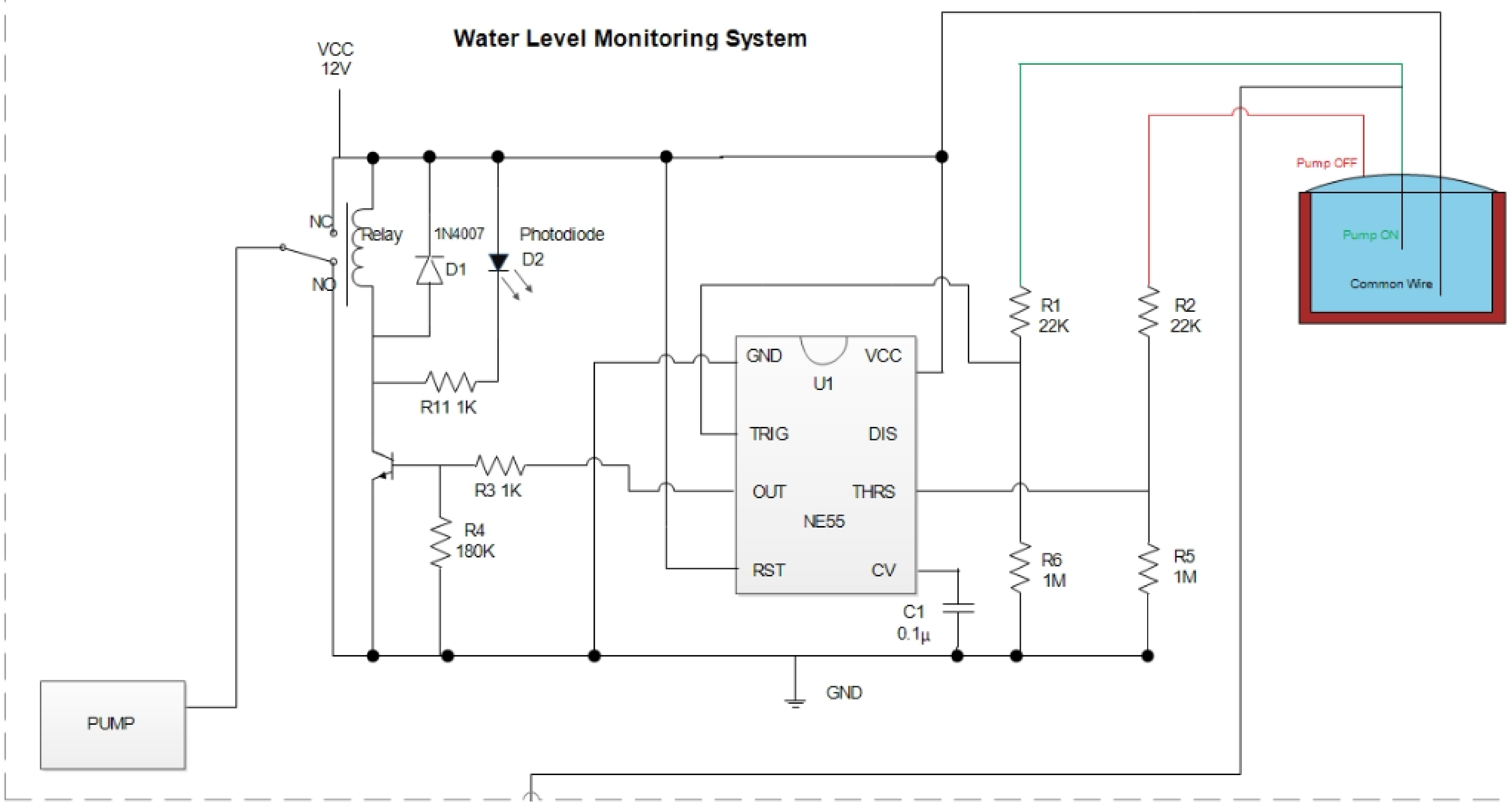
Design Method

- **Automatic Water Level Monitoring system:**
  The circuit developed for this block is entirely automatic and is used to monitor and regulate the water level of the tank, which acts as a water source for the irrigation setup. This circuit uses one NE 555 Timer IC, BC547 Transistor, SPDT Relay, and other passive components.

- **Soil Moisture Testing Unit :**
  This circuit is used to monitor the moisture level of the soil and supply water to it from the tank to meet the required moisture content of the soil. The circuits consist of one LM741 Opamp (that is used as a comparator), BC547 transistor, relay, and required passive components.

- Additionally, a water level-controlled switch has been added to connect the Water Level Monitoring Block and Moisture Level Tester. This ensures the irrigation setup is turned on only when there is sufficient water in the tank. This is achieved by using an Operational Amplifier as Comparator Switch.

Working: Automatic Water level monitoring system

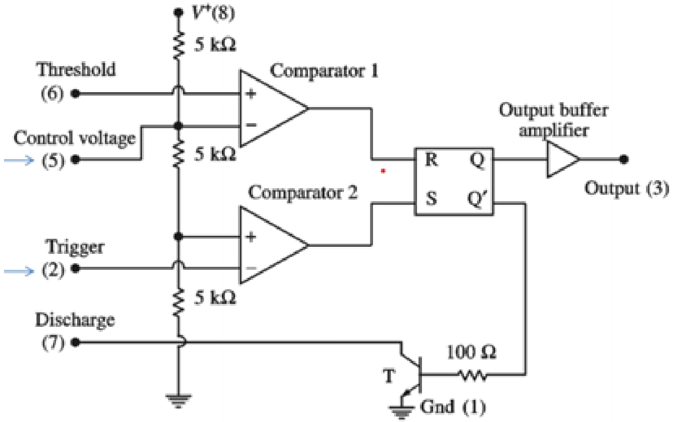


Working: Automatic Water level monitoring system

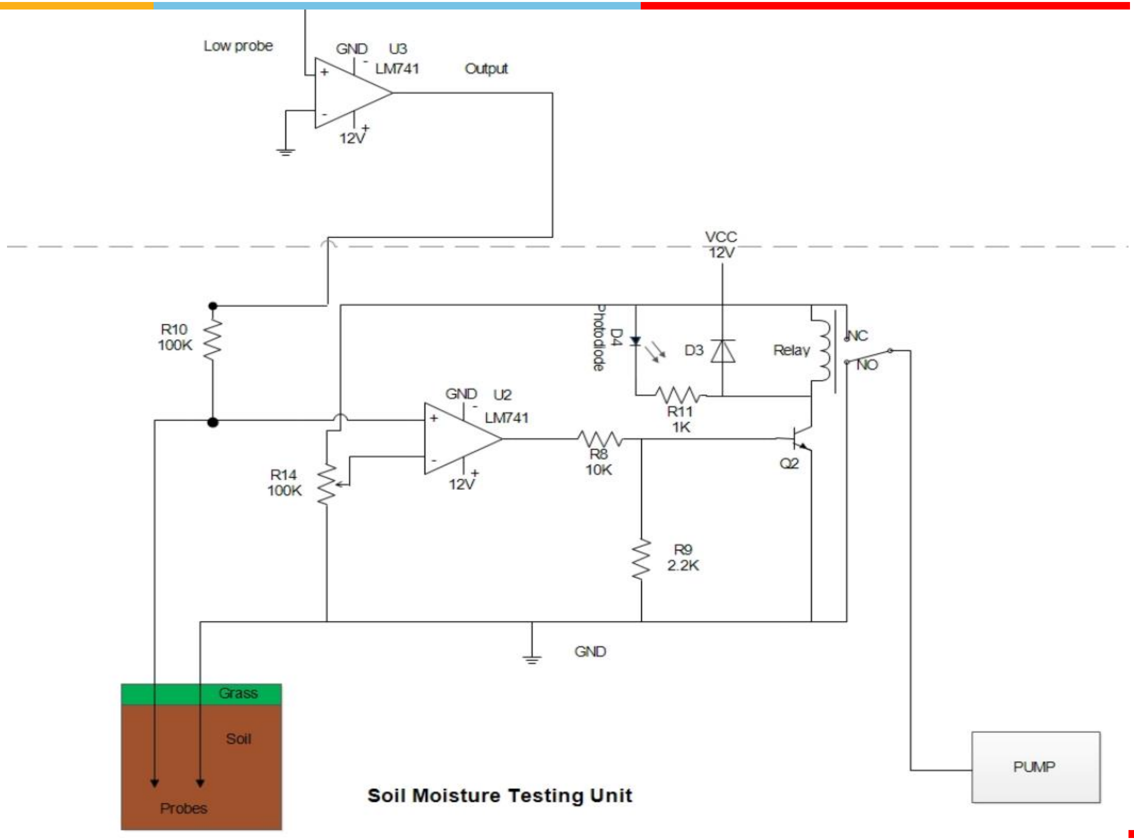
- **Case 1:** When the water level is below the low probe, the drop across the Trigger Pin and the Threshold Pin is zero, driving the comparator outputs ‘low’ and ‘high. This makes the output of the Timer IC high, which is further used to drive the relay in ON state.

- **Case 2:** When water level is above lower probe and below higher probe, the drop across Trigger pin becomes greater than VCC/3, driving the comparator outputs ‘low’. This makes the output of the Timer IC to remain unchanged.

- **Case 3:** When the water level reaches the upper wire contact, the internal RS flip-flop of IC is set and the output goes low. This low output at pin 3 switches off the transistor and de-energizes relay and disconnects the motor power supply.



Working: Soil moisture testing unit



Soil Moisture Testing Unit

Working: Soil moisture testing unit

The main principle used here is that when the soil is dry, it provides more resistance and when it is wet, the resistance offered is much less.

The potentiometer on the inverting terminal is calibrated to achieve a particular amount of moisture content in the soil.

- When the **soil moisture content is high,** it produces lower resistance compared to the potentiometer resistance on the inverting terminal. In this case –Vsat equals to 0. Thus, the output of the comparator is 0, indicating that the soil is wet. In this case, the photodiode does not glow, also the input of the BJT becomes 0, thus the relay will not be activated, and the **pump will be off.**

- When the **soil moisture content is low**, it produces higher resistance than the one set in the potentiometer on the inverting terminal. Therefore, the output of the comparator is set to +Vsat. Thus, the output of the comparator is 12V, indicating that the soil is dry. This makes the photodiode glow. The input of the BJT also becomes high, as a result current flows through it and the relay circuit is triggered, and the **pump is switched on**.

Simulation Results

Case 1: The soil is dry and the water tank is empty.

In this case the soil is dry and the tank is empty. So initially no water is being supplied to the soil. Then the pump associated with the water tank get triggered by the relay and the water level starts rising, simultaneously water is being supplied to the soil.

Case 2: The soil is dry and the water tank is full.

In this case since the soil is dry and the tank is full, therefore the pump associated with the soil moisture unit is triggered by the relay and water is being supplied to the soil.

Case 3: The soil is wet and the water tank is empty.

In this case, the output of U2 Opamp is low which switches off the pump 2 but the the tank is filled by switching on the pump 1.

Case 4: The soil is wet and the water tank is full.

Since all the conditions are met so the whole circuit will be in the off state i.e., both the pumps will be off.

Application/Advantages

- It can be used in applications like household purposes and farming purposes (irrigation) in an efficient and an automated way.
- The soil moisture testing unit can be used to maintain the moisture content and water the plants when required.
- In many places, it is not possible to manually operate the pump. The water level controller makes this process automatic.
- Low maintenance, Compact and Fully Automatic design.
- It saves water, energy and helps in maintaining soil quality.

Conclusion

- The project can be used to maintain a desired level of soil moisture via the means of a water tank. It also ensures that the water level of the tank does not cross a maximum and a minimum level.

- A water controlled switch has been used which ensures that the water irrigation setup is turned on only when there is sufficient water in the tank, making it efficient and is achieved by using an Operational amplifier as comparator switch.

- This circuit can be used for operating a 12V single phase pump. For large scale applications, the circuit components will have to be scaled accordingly.

- We were thus able to build an automated irrigation system using LM741 Operational Amplifier and IC555.

BITS Pilani
K K Birla Goa Campus

Thank You